

PRAJWAL KUMAR K

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EDUCATION

Indian Institute of Information Technology, Kottayam

Kottayam, Kerala

B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Aug 2024 – May 2028 (expected)

RESEARCH INTERESTS

Algorithms and systems, game theory, machine learning and data science, AI safety, and NLP.

SKILLS

Programming: Python, C++, Rust, MATLAB

ML and data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

Research tools: Jupyter, Simulink, FastAPI, Next.js, SQL

Current study: ISLP, Mathematics for Machine Learning, Stanford CS231n (ongoing)

Languages: Fluent in English, Kannada, German and Hindi.

EXPERIENCE AND VOLUNTEERING

Technical Lead, Openverse (Open Source Club)

May 2025 – Present

IIIT Kottayam

Kottayam, Kerala

– Lead technical work and PR reviews; contribute sorting visualizations to the team-built Algorithmia platform.

Coordinator, Coders Club

Jan 2025 – Present

IIIT Kottayam

Kottayam, Kerala

– Curate problems, configure Codeforces Gym rounds and help run campus programming contests.

RESEARCH AND SELECTED PROJECTS

PokeForge | *Gen 9 OU Battle Agents, Python, PyTorch*

GitHub | In progress

– Reproduced pinned FoulPlay and Metamon baselines in isolated simulator environments; testing simulator conformance and agent strength before drawing performance conclusions.

Bluff | *Repeated-Play Opponent Modeling*

GitHub

– Built an imperfect-information card-game simulator where agents lie and call bluffs; persistent Bayesian profiles model recurring opponents across rounds.

– Across eight synthetic profiles, removing the learned bluff detector reduced draw-aware score by 4.9–5.8 percentage points; this result is limited to the simulator and tested roster.

Drishti-XAI | *Retinal Imaging, MATLAB, Python*

SIH 2026 submission

– Built the MATLAB/Python prototype and evaluation pipeline for explainable retinal screening, covering image-quality checks, grading, segmentation, Grad-CAM, calibration and reporting.

– At a fixed model/threshold, internal APTOS (n=733) achieved 0.908 sensitivity/0.877 specificity; external Messidor-2 achieved 0.363/0.865 with AUC 0.666, exposing a transfer gap.

KPlaceNet | *Image Geolocation, PyTorch, ResNet-50*

GitHub

– Evaluated quadtree-based CNN geolocation; the best of 12 ResNet-50 configurations placed 3.33% of 3,000 spatially separated OSV-5M test images within 200 km.

PUBLICATION, ACHIEVEMENTS AND OPEN SOURCE

- First-listed author and conference presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025, pp. 1390–1395. DOI.
- Algorithmia selected for OOSC 4.0; team lightning talk at OOSC 3.0 and presentation at T4E 2025, IIT Madras.
- Best Software Award at IEEE MACE .hack(); 2026, with Team Hogorithm for Anchor.
- Open source: PettingZoo Leduc Hold'em (PR #1500, open); migrated the environment to an OpenSpiel-backed Shimmy adapter; focused tests passed.